SÈRIE 2

Exercici 1

Q1 d

Q2 b

Q3 a

Q4 c

Q5 a

Exercici 2

a)

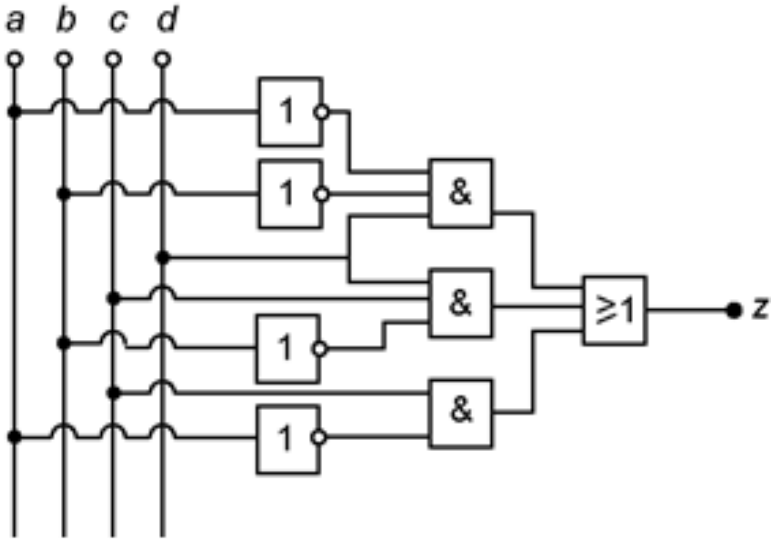
a	b	c	d	z
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

b)

$$z = \left(\bar{a}\,\bar{b}\,\bar{c}\,d\right) + \left(\bar{a}\,\bar{b}\,c\,\bar{d}\right) + \left(\bar{a}\,\bar{b}\,c\,d\right) + \left(\bar{a}\,b\,c\,\bar{d}\right) + \left(\bar{a}\,b\,c\,d\right) + \left(a\,\bar{b}\,c\,d\right)$$

simplificant: $z = \bar{a}\,\bar{b}\,d + \bar{b}\,c\,d + \bar{a}\,c$

c)



Exercici 3

a)

$$E_{\text{cons}} = P_1\, t_1 + P_2\left(t - t_1\right) = 1250\,\text{Wh} = 4500\,\text{kJ}$$

b)

$$c_{\text{r}} = \frac{P_1\, t_1}{E_{\text{cons}}} \cdot 100 = 80\,\%$$

c)

$$c_{\text{punta}} = E_{cons}\, p_{punta} = 0,4287\,€$$

$$c_{\text{vall}} = E_{cons}\, p_{vall} = 0,2712\,€$$

d)

$$e_q = \left(c_{\text{punta}} - c_{\text{vall}}\right) n\ 12 = 18,90\,€$$

Exercici 4

a)

$$\alpha = \frac{\omega_f - \omega_0}{t} = \frac{-n_0 \frac{2\pi}{60}}{t} = -8,727\,\text{rad/s}^2$$

b)

$$\varphi = \varphi_0 + \omega_0 t + \frac{1}{2}\alpha t^2 = 15,71 \times 10^3\,\text{rad}$$

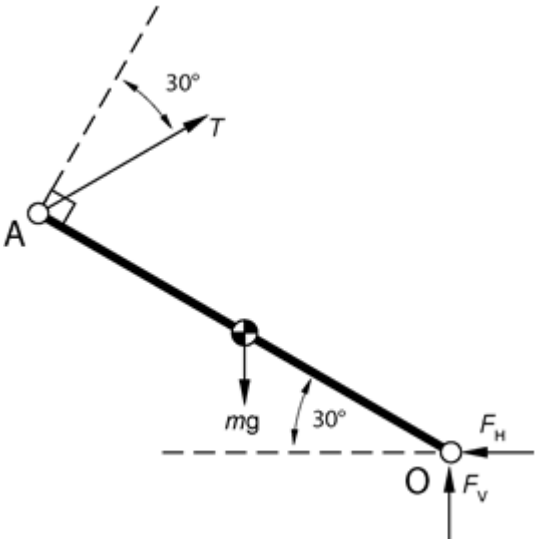
$$n = \frac{\varphi}{2\pi} = 2500 \text{ voltes}$$

c)

$$E_{diss} = \frac{1}{2} I \left(\omega_0^2 - \omega_f^2 \right) = 123,4\,\text{kJ}$$

Exercici 5

a)



b)

$$\sum M(\text{O}) = 0 \quad \rightarrow \quad T\cos(30)2L - m\,\text{g}\cos(30)L = 0 \quad \rightarrow T = \frac{m\,g}{2} = 245,2\ \text{N}$$

c)

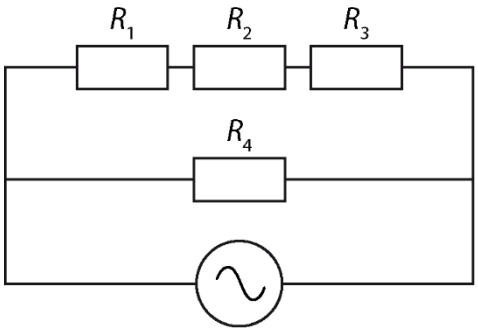
$$\left.\begin{array}{lll} \sum F_{\text{horitzontals}} = 0 & \rightarrow & T\cos(30) - F_{\text{H}} = 0 \\ \sum F_{\text{verticals}} = 0 & \rightarrow & T\sin(30) - mg + F_{\text{V}} = 0 \end{array}\right\} \begin{array}{l} F_{\text{H}} = T\sqrt{3}/2 = 212{,}3\ \text{N} \\ F_{V} = 3mg/4 = 367{,}8\ \text{N} \end{array}$$

d)

$$\Gamma = T\frac{d}{2} = 55{,}16\ \text{Nm}$$

Exercici 6

a)



b)

$$R_{\text{eq}} = \frac{1}{\dfrac{1}{R_1 + R_2 + R_3} + \dfrac{1}{R_4}} = 37{,}5\ \Omega$$

c)

$$I = \frac{U}{R_{\text{eq}}} = 6{,}133\ \text{A}; \quad P = \frac{U^2}{R_{\text{eq}}} = 1{,}411\ \text{kW}$$